2.6.4 Pharmacokinetics Written Summary

PrP-lowering divalent siRNA 2439-exNA

Other names: 2439-exNA, 2439-s4, PrP-siRNA

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LIST OF ABBREVIATIONS

AEX	anion exchange
CSF	cerebrospinal fluid
di-siRNA	divalent short interfering RNA
exNA	extended nucleic acid
GLP	Good Laboratory Practices
HPLC	high performance liquid chromatography
ICV	intracerebroventricular
IT	intrathecal
PNA	peptide nucleic acid
siRNA	short interfering RNA
TK	toxicokinetic

2.6.4 PHARMACOKINETICS WRITTEN SUMMARY

2.6.4.1 Brief Summary

PrP lowering divalent 2439-exNA (hereafter referred to as 2439-exNA) is a fully chemically modified oligonucleotide drug candidate incorporating extended nucleic acid and a 3' antisense tail unmatched to the RNA target that is intended for the treatment of human prion disease.

Single dose 2439-exNA toxicokinetic (TK) evaluations were conducted in the rat and dog. 2439-exNA was administered via intrathecal (IT) injection in Good Laboratory Practice (GLP) toxicology studies. In addition to the plasma toxicokinetic evaluation, tissue and CSF concentrations were also measured. The TK studies are summarized in Table 1. At the time of this IND submission, CSF concentrations are available for dogs and rats, and tissue concentrations are available for dogs, but analysis of rat tissue samples is ongoing.

Across single-dose studies, 2439-exNA demonstrated a profile which is consistent with IT administration of siRNA with evidence of uptake into the brain and liver tissue, and excretion via the kidneys. Following IT dose administration, the plasma concentration-time profiles for 2439-exNA males and females showed an apparent monophasic decline. T_{max} occurred at either the first or second target collection time of 0.5 or 4.00 hr in dogs and at the first collection time of 0.5 hours in rats.

2439-exNA was detected in the CSF for most dogs and all rats on Day 2 (~24 hours) following 2439-exNA administration, although the concentrations were highly variable. However, even in animals where 2439-exNA was not detected in CSF, exposures in spinal cord and brain confirmed that 2439-exNA had been delivered to the intrathecal space.

Evaluation of 2439-exNA tissues levels showed the highest concentration on Day 2 in the liver. 2439-exNA was observed in the lumbar and cervical spinal cord, with higher concentrations observed in the lumbar spinal cord which correlates to the site of administration. In the brain, 2439-exNA was observed at the highest concentration in the frontal cortex, hippocampus, and cerebellar cortex, but widespread exposure was noted with lower concentrations noted in the thalamus, putamen, and caudate. Generally, 2439-exNA remained present in liver, kidney, spinal cord, and brain at Day 29, but at lower concentrations compared to Day 2.

Based on in vitro evaluations, 2439-exNA has low potential as a substrate or inhibitor of metabolizing enzymes and membrane transporters.

In summary, the nonclinical pharmacokinetic studies support the administration of 2439-exNA to prion disease patients in a single dose clinical trial.

Table 1. Toxicokinetic Studies with 2439-exNA

Study Number	Study Description	GLP Status	Route	Dose (mg/kg)
A-703-01[a]	Single dose study with 4-week recovery in rats	GLP	IT	0.3, 1, and 3 mg

Study Number	Study Description	GLP Status	Route	Dose (mg/kg)
A-703-02 [a]	Single dose study with 4-week recovery in dogs	GLP	IT	20, 60, and 200 mg

[a] Plasma, CSF and tissue toxicokinetics assessed on the study; tissue TK was non-GLP

2.6.4.2 Methods of Analysis

To support the characterization of the disposition profile of 2439-exNA analytical methods were developed and validated to quantify 2439-exNA concentrations in rat and dog plasma, as described in Table 2. Qualified, fit for purpose analytical methods were also developed to analyze 2439-exNA concentrations in CSF and tissues.

Table 1: Methods for Toxicokinetic Analysis of 2439-exNA in Rats and Dogs

Species	Matrix	Quantitation Range	Validation Report Number	BA Study Report	TK Study Number
SD rats	Plasma	1 – 1,000 ng/mL	AN403-0-24	AN410.0-24	A-703-01
Beagle dog	Plasma	1 – 1,000 ng/mL	AN402.0-24	AN411.0-24	A-703-02
SD rats [a]	CSF	1 – 1,000 ng/mL	AN405.0-24	AN408.0-24	A-703-01
Beagle dog	CSF	1 – 1,000 ng/mL	AN404.0-24	AN409.0-24	A-703-02
SD rats [a]	Tissue	1 – 1,000 ng/mL	AN407.0-24	AN422.0-24	A-703-01
Beagle dog [a]	Tissue	20-10,000 ng/g	AN406.0-24	AN423.0-24	A-703-02

[a] Qualified method rather than validated method used

2.6.4.2.1 Measurement of 2439-exNA in plasma

The TK plasma assays employed bioanalytical evaluation of 2439-exNA by a hybridization-based Peptide Nucleic Acid Anion-Exchange High-Performance Liquid Chromatography (PNA-AEX-HPLC) method coupled to a fluorescence detector.

Long-term stability is currently on-going.

2.6.4.2.2 Measurement of 2439-exNA in CSF

The bioanalytical evaluation of 2439-exNA in CSF samples by a hybridization-based PNA-AEX-HPLC method coupled to a fluorescence detector.

2.6.4.2.3 Measurement of 2439-exNA in Tissues

The bioanalytical evaluation of 2439-exNA in tissues by an Anion-Exchange High-Performance Liquid Chromatography (AEX-HPLC) method with fluorescence detection that allows the sensitive and specific detection of the antisense strands of 2439-exNA in tissue samples.

The assay is based on the specific hybridization of a 13-mer PNA probe conjugated at the N-terminus with an Atto425 dye that is complementary to the antisense strands of the divalent 2439-exNA. In the first step

of the assay, dog tissue lysates are prepared by Proteinase K treatment. Then, the antisense strands of 2439-exNA will be hybridized with the PNA and subsequently analyzed by AEX-HPLC coupled to a fluorescence detector.

Quantification will be performed based on an external calibration curve generated from a standard dilution series in dog brain lysates. Linear calibration curves (weighted 1/X) will be calculated from 20 to 10,000 ng/g tissue.

Brain is the tissue of main concern and will therefore be tested more thoroughly than the secondary matrices liver and kidney, in which only selectivity, accuracy, precision and matrix effect will be assessed

2.6.4.3 Absorption

2.6.4.3.1 Single Dose Plasma Toxicokinetic Studies

2.6.4.3.1.1 Toxicokinetics of 2439-exNA in Rats (Study No.A-703-01)

Single dose TK of 2439-exNA was evaluated at three dose levels of 0.3, 1, and 3 mg following administration of a single IT dose to male and female rats. All animals in the study treated with 2439-exNA were systemically exposed to 2439-exNA. TK parameters are summarized in Table 3.

Table 3. TK Parameters for siRNA 2439-exNA Following Intrathecal Dose Administration in Rats

Dose (mg)	Sex	C_{max} (ng/mL)	T_{max} (hr)	Elimination Half-Life (hr)	Apparent Clearance (mL/hr)	Apparent Volume of Distribution (mL)	AUC_{last} (hr*ng/mL)	AUC_{∞} (hr*ng/mL)
0.3	Male	3500 (3500)	0.500 (0.500)	21.5 (1.70)	48.0 (49.1)	1490 (121)	7620 (7460)	7650 (7470)
	Female	2880	0.500	1.57	61.5	139	5970	5970
1	Male	13900 (13900)	0.500 (0.500)	14.3 (3.27)	43.1 (43.2)	889 (204)	28500 (28400)	28600 (28400)
	Female	13200 (13200)	0.500 (0.500)	10.7 (3.04)	43.7 (43.9)	678 (193)	28100 (28000)	28100 (28000)
3	Male	28300 (28300)	0.500 (0.500)	72.7 (3.37)	45.6 (46.1)	4790 (224)	76000 (75700)	76500 (75700)
	Female	28000 (28000)	0.500 (0.500)	6.99 (2.30)	41.1 (41.3)	415 (137)	84800 (84500)	84900 (84600)

The kinetic and systemic exposure parameters evaluated for 2439-exNA were C_{max}, T_{max}, elimination half-life, apparent clearance, apparent volume of distribution, AUC_{last}, and AUC_{∞}.

T_{max} occurred at the first target collection time of 0.5 for all groups. Elimination half-life and apparent volume of distribution values were variable. However, following reanalysis elimination half-life ranged from 1.57 to 3.37 hr, apparent clearance ranged from 41.3 to 61.5 mL/hr, and apparent volume of distribution ranged from 121 to 224 mL with no apparent sex or dose effect.

Systemic exposure, as measured by group mean C_{max}, AUC_{last}, and AUC_{∞}, was similar, within 2-fold, when comparing males and females, both before and after reanalysis. C_{max}, AUC_{last}, and AUC_{∞} increased a near dose proportional manner as the dose was increased by 9.5-fold, from 0.367 to 3.49 mg. Following reanalysis, C_{max}, AUC_{last}, and AUC_{∞} increased by 8.1-, 9.9-, and 9.9-fold, respectively for males and by 9.7-, 14.2-, and 14.2-fold, respectively for females.

2.6.4.3.1.2 Toxicokinetics of 2439-exNA in Dogs (Study No.A-703-02)

Single dose TK of 2439-exNA was evaluated at three dose levels of 20, 60, and 200 mg following administration of a single IT dose to male and female Beagle dogs. All animals in the study treated with 2439-exNA were systemically exposed to 2439-exNA. TK parameters are presented in Table 4

Table 4. TK Parameters for siRNA 2439-exNA Following Intrathecal Dose Administration in Dogs

Dose (mg)	Sex	Animal ID	Cmax (ng/mL)	Tmax (hr)	Elimination Half-Life (hr)	Apparent Clearance (mL/hr)	Apparent Volume of Distrib-ution (mL)	AUClast (hr*ng/mL)	AUC∞ (hr*ng/mL)
20	Male	201	212	0.500	1.67	15000	36300	1310	1330
		202	143	4.00	ND	ND	ND	950	ND
		203	291	0.500	1.75	17200	43500	1150	1160
		204	145	4.00	3.59	14700	76200	1340	1360
		205	198	4.00	2.42	11800	41400	1680	1690
		206	412	0.500	1.48	12000	25600	1650	1660
		Mean	**233**	**2.25**	**2.18**	**14200**	**44600**	**1350**	**1440**
		SE	**42**	**0.78**	**0.39**	**1000**	**8500**	**120**	**100**
	Female	251	311	0.500	1.79	17100	44200	1160	1170
		252	289	0.500	2.67	11900	45700	1680	1680
		253	832	0.500	2.10	8460	25600	2340	2370
		254	292	4.00	ND	ND	ND	1720	ND
		255	352	0.500	1.60	13300	30700	1490	1500
		256	265	0.500	1.76	15600	39600	1270	1280
		Mean	**390**	**1.08**	**1.98**	**13300**	**37200**	**1610**	**1600**
		SE	**89**	**0.58**	**0.19**	**1500**	**3900**	**170**	**210**
60	Male	301	3500	0.500	1.26	4900	8940	12200	12300
		302	1080	4.00	4.71	8010	54400	7320	7490
		303	4870	0.500	1.20	3530	6100	17000	17000
		304	3940	0.500	1.45	4180	8740	14300	14300
		305	4610	0.500	2.38	2820	9660	21300	21300
		306	3590	0.500	2.01	3780	11000	15900	15900
		Mean	**3600**	**1.08**	**2.17**	**4540**	**16500**	**14700**	**14700**
		SE	**550**	**0.58**	**0.54**	**750**	**7600**	**1900**	**1900**
		351	4960	0.500	1.90	3020	8280	19800	19800
		352	1870	4.00	3.42	4960	24500	12000	12100
		353	9660	0.500	2.14	2020	6250	29700	29700
		354	9280	0.500	1.15	2050	3390	29300	29300
		355	3010	4.00	2.73	3260	12800	18400	18400

	Female	356	2030 (2030)	4.00 (4.00)	4.83 (2.21)	5560 (5580)	38700 (17800)	10800 (10700)	10800 (10700)
		Mean	**5130 (5130)**	**2.25 (2.25)**	**2.69 (2.26)**	**3480 (3480)**	**15700 (12200)**	**20000 (20000)**	**20000 (20000)**
		SE	**1440 (1440)**	**0.78 (0.78)**	**0.53 (0.31)**	**610 (610)**	**5500 (3200)**	**3300 (3300)**	**3300 (3300)**
200	Male	401	5530	0.500	2.78	3830	15400	52000	52200
		402	11600	0.500	1.74	3100	7760	64600	64600
		403	11600	4.00	1.83	2630	6940	76100	76100
		404	9750	4.00	1.69	2460	5980	81400	81400
		405	10300	4.00	3.19	2060	9480	97000	97000
		406	12900 (12900)	4.00 (4.00)	37.7 (1.55)	2220 (2220)	121000 (4990)	90000 (90000)	90100 (90000)
		Mean	**10300 (10300)**	**2.83 (2.83)**	**8.16 (2.13)**	**2710 (2710)**	**27700 (8430)**	**76900 (76900)**	**76900 (76900)**
		SE	**1100 (1100)**	**0.74 (0.74)**	**5.92 (0.28)**	**270 (270)**	**18700 (1500)**	**6800 (6800)**	**6700 (6700)**
	Female	451	25400	0.500	1.76	1590	4040	126000	126000
		452	17600	4.00	1.91	1810	5000	110000	110000
		453	7410	4.00	2.49	2700	9700	73800	74000
		454	7800	4.00	3.95	2720	15500	72000	73500
		455	14400 (14400)	4.00 (4.00)	25.9 (1.41)	1580 (1590)	59100 (3230)	126000 (126000)	126000 (126000)
		456	26700 (26700)	0.500 (0.500)	5.10 (4.65)	1530 (1530)	11300 (10300)	130000 (130000)	130000 (130000)
		Mean	**16500 (16500)**	**2.83 (2.83)**	**6.85 (2.70)**	**1990 (1990)**	**17400 (7960)**	**106000 (106000)**	**107000 (107000)**
		SE	**3400 (3400)**	**0.74 (0.74)**	**3.84 (0.54)**	**230 (230)**	**8500 (1930)**	**11000 (11000)**	**11000 (11000)**

All Vehicle Control and pre-dose test article plasma samples were below the limit of quantitation for 2439-exNA. Concentrations of 2439-exNA were present in the plasma in a dose dependent trend for the 20, 60, and 200 mg 2439-exNA groups.

Following IT dose administration, the plasma concentration-time profiles for the 20, 60, and 200 mg 2439-exNA males and females showed an apparent monophasic decline.

T_{max} occurred at either the first or second target collection time of 0.5 or 4.00 hr for all animals. Group mean elimination half-life, apparent clearance, and apparent volume of distribution values were similar,

within 2-fold, when comparing males and female for a given dose. When comparing the three different dose amounts, elimination half-life was similar across the dose levels, however apparent clearance and apparent volume of distribution decreased as the dose was increased for a given sex.

Systemic exposure, as measured by group mean C_{max}, AUC_{last}, and AUC_{∞}, was similar, within 2-fold, when comparing males and females. C_{max}, AUC_{last}, and AUC_{∞} increased in an apparent greater than dose proportional manner as the dose was increased by approximately 10-fold, from 20 to 200 mg. C_{max}, AUC_{last}, and AUC_{∞} increased by 44.2-, 57.0-, and 53.4-fold, respectively for males and by 42.3-, 65.8-, and 66.9-fold, respectively for females. However, when evaluating the approximately 3.3-fold increase from 60 to 200 mg, systemic exposure increased closer to dose proportionally. C_{max}, AUC_{last}, and AUC_{∞} increased by 2.9-, 5.2-, and 5.2-fold, respectively for males and by 3.2-, 5.3-, and 5.4-fold, respectively for females. The greater than dose proportional increase was most pronounced as the dose was increased approximately 3-fold from 20 to 60 mg. C_{max}, AUC_{last}, and AUC_{∞} increased by 15.5-, 10.9-, and 10.2-fold, respectively for males and by 13.2-, 12.4-, and 12.5-fold, respectively for females.

2.6.4.4 Distribution

2.6.4.4.1 CSF Toxicokinetics

2.6.4.4.1.1 CSF Toxicokinetics of 2439-exNA in Rats (Study No.A-703-01)

The concentration of 2439-exNA was measured (non-GLP) in the CSF of male and female rats on Days 1 (~24 hours) and 28 after a single IT injection of 0.3, 1, or 3 mg of 2439-exNA. The results are summarized in Table 5.

Table 5. Mean CSF Concentrations of siRNA 2439-exNA Following Intrathecal Dose Administration in Rats

		Core			Recovery		
Dose (mg)	**Sex**	**Mean CSF Concentration (ng/mL)**	**SD**	**n**	**Mean CSF Concentration (ng/mL)**	**SD**	**n**
0.3	Male	33.8	34.7	5	2.1	0.8	3
	Female	128.4	321.8	9	2.8	1.0	4
1	Male	235.3	301.9	10	5.6	2.9	5
	Female	397.8	581.4	9	8.2	3.7	4
3	Male	3323.1	2839.9	8	7.9	7.4	4
	Female	1248.0	2879.4	9	11.3	8.1	5

All control group samples were below the limit of quantitation for 2439-exNA. Mean CSF concentrations of 2439-exNA increased as the dose was increased. This dose response was more apparent on Day 2 than on Day 29. Mean CSF concentrations from recovery animals were markedly lower compared to core animals for a given dose.

2.6.4.4.1.2 CSF Toxicokinetics of siRNA 2439-exNA in Dogs (Study No.A-703-02)

The concentration of 2439-exNA was measured (non-GLP) in the CSF male and female dogs on Day 1 (~24 hours) and 28 after a single IT injection of 20, 60, and 200 mg of 2439-exNA. The results are summarized in Table 6.

Table 6. Mean CSF Concentrations of siRNA 2439-exNA Following Intrathecal Dose Administration in Dogs

		Core			Recovery		
Dose (mg)	**Sex**	**Mean CSF Concentration (ng/mL)**	**SD**	**n**	**Mean CSF Concentration (ng/mL)**	**SD**	**n**
20	Male	1729.9	2989.7	3	3.1	2.3	2
	Female	179.9	310.8	4	1.6	N/A	1
60	Male	26582.2	52679.7	4	2.2	0.8	2
	Female	11250.9	15909.5	2	6.2	0.6	2
200	Male	54175.7	105654.7	4	13.5	N/A	1
	Female	90616.2	136139.4	4	45.1	N/A	1

n = number of animals with measurable CSF concentrations; does not include animals with BLOQ values. Core groups had 4 total animals/sex. Recovery groups had 2 total animals/sex.

All Vehicle Control CSF samples were below the limit of quantitation for 2439-exNA. Generally, 2439-exNA was detected in the CSF for most dogs in the 20, 60, and 200 mg siRNA 2439-exNA groups, with higher concentrations of 2439-exNA observed in the CSF on Day 2 than on Day 29.

2.6.4.4.2 Tissue Toxicokinetics

2.6.4.4.2.1 Tissue Toxicokinetics of 2439-exNA in Rats (Study No.A-703-01)

The concentration of 2439-exNA was measured (non-GLP) in the spinal cord, brain (multiple regions), liver and kidney of male and female rats on Day 1 (~24 hours) and 28 after a single IT injection of 0.3, 1, and 3 mg following administration of 2439-exNA. The concentration analysis of the rat tissues is ongoing.

2.6.4.4.2.2 Tissue Toxicokinetics of s 2439-exNA in Dogs (Study No.A-703-02)

The concentration of 2439-exNA was measured (non-GLP) in the spinal cord, brain (multiple regions), liver and kidney of male and female dogs on Day 1 (~24 hours) and 28 after a single IT injection of 20, 60, and 200 mg following administration of 2439-exNA. The results are summarized in Table 7.

Table 7. Mean Tissue Concentrations of siRNA 2439-exNA Following Intrathecal Dose Administration in Dogs

			Core			Recovery		
Tissue	**Dose (mg)**	**Sex**	**Mean Concentration (ng/mL)**	**SD**	**n**	**Mean Concentration (ng/mL)**	**SD**	**n**
Spinal Cord, cervical	20	Male	2521.7	3649.4	4	3009.8	1399.9	2
		Female	4991.65	4068.2	4	1944.3	2583.3	2
	60	Male	10289.7	9288.0	4	4573.0	19.9	2
		Female	6699.9	10927.5	4	10539.2	938.8	2
	200	Male	18514.9	30473.1	4	9870.9	13353.0	2
		Female	37485.1	37836.5	4	9816.6	13705.1	2
Spinal Cord, lumbar	20	Male	9814.6	10681.6	4	12,688.2	1993.3	2
		Female	8883.5	6678.5	4	2,144.1	2557.5	2

	60	Male	32922.5	25363.8	4	9,264.8	3014.6	2
		Female	17311.1	29251.1	4	58,221.0	17052.3	2
	200	Male	68965.9	97825.1	4	17,066.1	22483.4	2
		Female	129316.4	178096.1	4	24,503.5	31585.5	2
Brain, Frontal Cortex	20	Male	5808.1	8687.2	3	5,564.8	3,030.9	2
		Female	3,083.5	3,420.7	4	30.5	13.9	2
	60	Male	18,050.2	18,545.2	4	13,637.8	3,341.2	2
		Female	5,939.8	11,759.9	4	19,153.1	6,331.7	2
	200	Male	24,870.0	47,001.8	4	27,978.3	39,402.3	2
		Female	32,878.2	37,075.9	4	13,782.7	19,334.6	2
Brain, Hippocampus	20	Male	1944.8	3116.8	3	4211.0	4558.8	2
		Female	1884.3	1295.6	4	680.0	933.0	2
	60	Male	11369.0	15792.5	4	1561.9	762.3	2
		Female	4570.2	8084.5	4	7840.7	24.8	2
	200	Male	14793.2	26770.9	4	11514.8	16227.8	2
		Female	31450.7	40847.6	4	12025.1	16909.5	2
Brain, Caudate	20	Male	100.5	104.4	3	743.3	969.8	2
		Female	87.3	98.3	4	25.5	N/A	1
	60	Male	401.7	431.8	4	259.0	192.1	2
		Female	151.8	251.7	4	307.0	65.1	2
	200	Male	5549.3	10694.2	4	412.6	395.8	2
		Female	1013.8	1138.6	4	262.8	293.2	2
Brain, Putamen	20	Male	143.6	148.4	2	316.2	359.5	2
		Female	61.6	47.5	4	34.9	N/A	1
	60	Male	236.8	251.3	3	N/A	N/A	0
		Female	38.0	18.1	3	13436.8	18493.7	2
	200	Male	6,090.9	11,818.3	4	3,490.2	4,867.1	2
		Female	4,377.6	4,989.0	4	1,287.3	1,749.3	2
Brain, Thalamus	20	Male	507.2	N/A	1	1273.3	1655.9	2
		Female	40.5	18.7	4	108.2	N/A	1
	60	Male	266.1	299.6	4	68.0	32.9	2
		Female	738.2	1420.3	4	372.3	91.8	2
	200	Male	2248.7	4240.0	4	1,817.3	2524.6	2
		Female	1413.6	1741.6	4	2,363.0	3226.2	2
Brain, Cerebellar Cortex	20	Male	9012.8	11043.7	3	12,842.1	14,786.0	2
		Female	6282.5	3491.2	4	1,988.5	2,512.4	2
	60	Male	22536.3	25346.7	4	13,193.7	4,386.8	2
		Female	11923.2	22902.4	4	17,387.0	23,688.5	2
	200	Male	34942.3	55057.5	4	20,839.6	28,867.1	2
		Female	57206.5	67652.1	4	31,566.4	44,070.6	2
Liver	20	Male	45,014.8	9,216.9	4	14,062.1	15,282.7	2
		Female	60,330.3	6,484.1	4	21,460.9	16,451.3	2
	60	Male	110,623.2	16,646.4	4	32,326.2	2,843.6	2
		Female	132,823.0	24,710.9	4	49,824.2	12,932.1	2
	200	Male	220,105.5	30,202.2	4	84,174.0	9,470.0	2
		Female	246,838.1	50,304.6	4	90,747.3	51,348.0	2
Kidney	20	Male	3,270.1	771.4	4	863.2	290.6	2

		Female	3,520.8	628.1	4	1,354.3	954.5	2
	60	Male	13,665.3	10,686.4	4	4,588.2	67.5	2
		Female	15,764.8	8,329.9	4	2540.4	1732.2	2
	200	Male	58,017.2	15,466.2	4	12,858.7	4,262.9	2
		Female	63,806.3	15,271.5	4	24,309.0	1,651.9	2

n = number of animals with measurable concentrations; does not include animals with BLOQ values.
Core groups had 4 total animals/sex. Recovery groups had 2 total animals/sex.

The highest concentration of 2439-exNA was present in the liver. 2439-exNA was observed in the lumbar and cervical spinal cord, with higher concentrations observed in the lumbar spinal cord which correlates to the site of administration. 2439-exNA was observed in the brain, with frontal cortex, hippocampus, and cerebellar cortex generally containing higher concentrations than the thalamus, putamen, and caudate. 2439-exNA was also observed in the kidney. Generally, 2439-exNA remained present in liver, kidney, spinal cord, and brain at Day 29, but at lower concentrations compared to Day 2. The concentration of 2439-exNA demonstrated a dose proportional trend.

2.6.4.5 Metabolism and Excretion

No metabolism or excretion studies were conducted or planned with 2439-exNA

2.6.4.7 Pharmacokinetic Drug Interactions

2.6.4.7.1 2439-exNA: Cytochrome P450 Induction in Cultured Human Hepatocytes (Study XT233111)

This purpose of this study was to evaluate the effect of 2439-exNA (0.4 to 400 nM) as an inducer of human CYP1A2, CYP2B6 and CYP3A4. Three cultures of cryopreserved primary human hepatocytes were treated with 2439-exNA (0.4 to 400 nM) to evaluate the effect on mRNA expression of human CYP1A2, CYP2B6 and CYP3A4.

Treatment of cultured human hepatocytes with up to 400 nM 2439-exNA resulted in normal hepatocyte morphology. Additionally, treatment of hepatocytes with up to 400 nM 2439-exNA had no effect (i.e., < 25% increase) on LDH release. There was no induction of CYP1A2, CYP2B6 or CYP3A4 mRNA levels (< 2-fold change) following treatment of cultured human hepatocytes with up to 400 nM 2439-exNA.

2.6.4.7.2 2439-exNA: Cytochrome P450 Inhibition in Human Hepatocytes (Study XT235113)

The purpose of this study was to evaluate the ability of 2439-exNA (0.4 to 400 nM) to inhibit CYP1A2, CYP2B6, CYP2C8, CYP2C9, CYP2C19, CYP2D6 and CYP3A4/5 (using two different substrates) in cryopreserved human hepatocytes by determining IC_{50} values.

A maximum of 35.3% inhibition of CYP2C8 by 2439-exNA was observed, with this maximum inhibition occurring at the second-highest dose tested. The IC_{50} value was reported as > 400 nM. 2439-exNA was not a direct inhibitor of CYP1A2, CYP2B6, CYP2C9, CYP2C19, CYP2D6, CYP3A4/5 (midazolam) or CYP3A4/5 (testosterone).

2439-exNA was not a time-dependent inhibitor of any CYP evaluated.

2.6.4.7.3 2439-exNA: ABC and SLC Transporter Inhibition and Substrate Potential in Cells (Study XT238083)

This purpose of this was to evaluate 2439-exNA as an inhibitor of the human transporters P-gp, BCRP, OATP1B1, OATP1B3, OAT1, OAT3, OCT2, MATE1 and MATE2-K, and a substrate of human transporters P-gp and BCRP.

A maximum of 27.2% inhibition of OAT3 was observed in the presence of 2439-exNA concentrations up to 400 nM. 2439-exNA was not an inhibitor of P-gp, BCRP, OATP1B1, OATP1B3, OAT1, OCT2, MATE1 or MATE2-K under the conditions examined. 2439-exNA was not a substrate of P-gp of BCRP under the conditions examined.

2.6.4.8 Other Pharmacokinetic Studies

No additional PK studies with 2439-exNA were performed.

2.6.4.9 Discussion and Conclusions

PrP lowering divalent 2439-exNA (hereafter referred to as 2439-exNA) is a fully chemically modified oligonucleotide drug candidate incorporating extended nucleic acid and a 3' antisense tail unmatched to the RNA target that is intended for the treatment of human prion disease.

Single dose 2439-exNA TK evaluations were conducted in the rat and dog. 2439-exNA was administered via IT injection in GLP toxicology studies. In addition to the plasma toxicokinetic evaluation, tissue and CSF concentrations were also measured. The rat and dog were used in the pharmacology and toxicology evaluations with similar drug formulations and the IT route of administration was selected because it is the proposed clinical route. At the time of this IND submission, CSF concentrations are available for dogs and rats, tissues concentrations are available for dogs, but analysis of rat tissue samples is ongoing.

Across single-dose studies, 2439-exNA demonstrated a profile which is consistent with IT administration of siRNA molecules with evidence of uptake into the brain and liver tissue, and excretion via the kidneys. Following IT dose administration, the plasma concentration-time profiles for 2439-exNA males and females showed an apparent monophasic decline. T_{max} occurred at either the first or second target collection time of 0.5 or 4.00 hr in dogs and at the first collection time of 0.5 hours in rats.

2439-exNA was detected in the CSF for most dogs and all rats on Day 2 (~24 hours) following 2439-exNA administration, although the concentrations were highly variable. However, even in dogs where 2439-exNA was not detected in CSF, exposures in spinal cord and brain confirmed that 2439-exNA was delivered to the intrathecal space.

Evaluation of 2439-exNA tissues levels in dogs showed that the highest concentration 2439-exNA was present in the liver. 2439-exNA was observed in the lumbar and cervical spinal cord, with higher concentrations observed in the lumbar spinal cord which correlates to the site of administration. 2439-exNA was observed in the brain, with frontal cortex, hippocampus, and cerebellar cortex generally containing higher concentrations than the thalamus, putamen, and caudate. 2439-exNA was also observed in the kidney. Generally, siRNA 2439-exNA remained present in liver, kidney, spinal cord, and brain at Day 29, but at lower concentrations compared to Day 2. The concentration of 2439-exNA demonstrated a dose proportional trend.

Based on in vitro evaluations, 2439-exNA has low potential as a substrate or inhibitor of metabolizing enzymes and membrane transporters Based on in vitro evaluations, 2439-exNA has low potential as a substrate or inhibitor of metabolizing enzymes and membrane transporters.

In summary, the nonclinical pharmacokinetic studies support the administration of 2439-exNA to prion disease patients in a single dose clinical trial.